

FFY2021 MAS Portfolio Summary

Overview

The FFY2021 ISWS CHAMP portfolio consists of **13 Mapping Activity Statements (MASs)** totaling approximately **\$3.62 million in FEMA funding**. The portfolio spans the full spectrum of Risk MAP activities: statewide program management (PM), community outreach and mitigation strategies (COMS), LOMR review, and 9 Flood Risk Projects (FRPs) across Illinois. The FRPs break into three broad categories: (1) near-final PMR/redelineation projects carrying existing county-level studies through database development to QR1 (Rock Island, Whiteside, Henry), (2) a project completing a Paper Inventory Reduction to effective (Stark), and (3) multi-phase data development projects for counties and watersheds where new H&H analysis is needed to produce new FIRM panels (Montgomery, Hamilton, Wabash, Bond, Kishwaukee). Many projects are explicit continuations of prior-year MASs, reflecting the multi-year lifecycle of DFIRM production. No project in this portfolio completes the full lifecycle to effective FIRM panels in FY21 alone—all FRPs will require at least one additional future MAS for Preliminary or Post-Preliminary Map Production.

All projects list **Glenn Heistand** as CTP Project Manager and **John Wethington** as FEMA Regional Project Officer. The LOMR project uses **Christopher R. Hanstad** as PM. Period of Performance for FRPs is 09/01/2021 – 09/30/2025; PM and COMS SOWs run 09/01/2021 – 09/30/2023.

Projects

ISWS-21-LOMR: LOMR Review Partners — Statewide Illinois

- **Activity Type:** LOMR Review Partners (CLOMR/LOMR Technical Review)
- **Phase:** Ongoing operational program; FY21 is a standalone 2-year grant
- **Scope:** ISWS assumes responsibility for reviewing and recommending approval of all CLOMR and LOMR requests across the entire State of Illinois. The CTP performs initial and final technical review, prepares draft determination documents, conducts QA/QC, handles fee administration, post-processes issued LOMRs, manages appeals and comment processing, and maintains MIP records. FEMA retains authority for final issuance. Levee-related cases are referred to FEMA's contractor. Key staff include Chris Hanstad (PM/Engineer), Brian Chaille, Jimmy Powell, Addison Jobe, Shelly Fuller (post-processing), Sam Chakravorty, and Greta Buckley (GIS).
- **Deliverables:** MIP case entries within 24 hours of receipt; acknowledgment letters; draft CLOMR/LOMR determination documents; QA/QC review documentation; post-processing database submittals; audit-ready case files.
- **Period:** August 1, 2021 – September 30, 2023
- **Budget:** \$370,000
- **Multi-year:** No (distinct operational program; each grant year is independently scoped). Program is ongoing across grant cycles.

- **Recommended Next Steps:** Continue LOMR review program with a follow-on FY23 MAS. Monitor for potential Physical Map Revision (PMR) triggers from significant LOMRs.
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ISWS 21-01: Program Management (PM) SOW — Statewide Illinois

- **Activity Type:** Program Management (PM SOW)
 - **Phase:** Annual PM grant; standalone 2-year administration period
 - **Scope:** ISWS CHAMP program management for FY21 Risk MAP grant activities. Key tasks include: joint Business Plan with IDNR/OWR (\$25,000), global program management and quarterly reporting of risk assessments and CTP performance metrics (\$25,000), outreach activities through IAFSM, ILGISA, ASFPM conferences and ISWS website (\$5,000), CNMS database updates and validation assessment incorporation (\$20,000), and Programmatic QA/QC Plan development (\$5,000). Staff certification maintenance (CFM, PE, GISP, AICP) is an explicit program management goal.
 - **Deliverables:** Joint Business Plan (3 months from award); Program Management Plan (4 months); Outreach meeting notices; CNMS spatial database updates (ongoing); Programmatic QA/QC Plan (5 months from award); quarterly performance reports.
 - **Period:** 09/01/2021 – 09/30/2023
 - **Budget:** \$80,000
 - **Multi-year:** N/A (PM SOWs are annual/biennial overhead; not multi-year in the project sense)
 - **Recommended Next Steps:** Prepare FY22/FY23 PM SOW. Finalize joint Business Plan with IDNR/OWR.
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ISWS 21-02: Rock Island County — Data Development, Redelineation & PMR through QR1

- **Activity Type:** Flood Risk Project — Data Development, Redelineation, FIRM Database (QR1)
- **Phase:** Continuation of ISWS 17-06 (FY17 data development); this MAS advances to QR1 FIRM database
- **Scope:** Advances the Rock Island County 48-panel PMR through QR1 FIRM database submittal. Work includes incorporation of new study data (Zone AE & Zone A) from prior MAS ISWS 17-06; development of additional H&H for Kyte Creek, Eckhart Creek, Coal Creek, Coal Creek Tributary, and Shaffer Creek using HEC-HMS and HEC-RAS (1D and 2D); a 2D model for the East Moline leveed area; redelineation of the Mississippi River Zone AE (37.7 mi), Mill Creek Zone AE (7.8 mi), and seven levee areas (Rock Island City, Arsenal, East Moline, Milan West/Big Island, Milan East, Milan South Slough, Meredosia-RI portion); and survey data acquisition via contract.
- **Deliverables:** Survey data; topographic data (2020 LiDAR); H&H reports with independent QA/QC; floodplain mapping spatial files; Draft FIRM Database through QR1; Project Outreach and Communication Plan; project management plan; QA/QC log.
- **Period:** 09/01/2021 – 09/30/2025
- **Budget:** \$770,000
- **Multi-year:** Yes. Explicitly builds on MAS ISWS 17-06. This MAS stops at QR1 (Draft FIRM Database); Preliminary Map Production and Post-Preliminary (QR2–QR8, CCO, due process, effective FIRM) are NOT funded in this MAS.

- **Recommended Next Steps:** After QR1 FIRM database passes, fund a follow-on MAS for Preliminary Map Production through QR2, then distribution and Post-Preliminary (CCO meeting, appeals, final effective FIRM). Resolve levee PAL/accreditation status for 7 levee areas before preliminary issuance.
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ISWS 21-03: Whiteside County — Redelineation & PMR through QR1

- **Activity Type:** Flood Risk Project — Redelineation, FIRM Database (QR1)
 - **Phase:** Continuation of ISWS 17-06 (FY17 data development); this MAS advances to QR1
 - **Scope:** Advances the Whiteside County 50-panel PMR through QR1 FIRM database submittal. Incorporates new study data (Zone AE & Zone A) from MAS ISWS 17-06; redelineates the Mississippi River Zone AE effective data on PMR panels (10.2 mi); redelineates the Meredosia Levee area (Whiteside County portion, 12.4 sq mi Zone AE). No new H&H analysis required—all existing data from prior MAS. Topographic data uses 2020 LiDAR (4 ppsm, NVA 0.196 m) from ISGS.
 - **Deliverables:** Topographic/terrain data; base map (2015 IDOT aerial); floodplain mapping spatial files; Draft FIRM Database through QR1; communication plan; QA/QC documentation.
 - **Period:** 09/01/2021 – 09/30/2025
 - **Budget:** \$150,000
 - **Multi-year:** Yes. Builds on ISWS 17-06. Stops at QR1; Preliminary Map Production and Post-Preliminary are NOT funded.
 - **Recommended Next Steps:** Fund Preliminary Map Production MAS after QR1. Resolve Meredosia Levee status. Monitor new Whiteside County LiDAR availability from ISGS for potential improvement to study quality before preliminary issuance.
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ISWS 21-04: Henry County — Redelineation & PMR through QR1

- **Activity Type:** Flood Risk Project — Redelineation, FIRM Database (QR1)
 - **Phase:** Continuation of ISWS 17-06 (FY17 data development); this MAS advances to QR1
 - **Scope:** Advances the Henry County 21-panel PMR through QR1 FIRM database submittal. Incorporates new study data (Zone AE & Zone A) from MAS ISWS 17-06; redelineates the Green River Zone AE effective data (6.0 mi). No new H&H analysis required—all from ISWS 17-06. Uses 2020 LiDAR from ISGS. Project footprint is limited (Cleveland, Coal Valley, Colona, and unincorporated Henry County).
 - **Deliverables:** Topographic data; base map; floodplain mapping (Green River 6.0 mi AE redelineation); Draft FIRM Database through QR1; communication plan; QA/QC documentation.
 - **Period:** 09/01/2021 – 09/30/2025
 - **Budget:** \$125,000
 - **Multi-year:** Yes. Builds on ISWS 17-06. Stops at QR1; Preliminary Map Production is NOT funded.
 - **Recommended Next Steps:** Fund Preliminary Map Production after QR1. This is the smallest of the three co-MAS 17-06 continuation projects; straightforward path to preliminary once QR1 passes.
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ISWS 21-05: Stark County — Preliminary & Post-Preliminary Map Production (Paper Inventory Reduction completion)

- **Activity Type:** Flood Risk Project — Preliminary Map Production, Post-Preliminary Map Production, Flood Risk Products, Paper Inventory Reduction to Effective
 - **Phase:** Final phase of Stark County Paper Inventory Reduction (prior MAS: ISWS 19-03)
 - **Scope:** Advances the countywide digital FIRM for Stark County (19 new panels) from QR2 through Effective, completing the Paper Inventory Reduction project that began under ISWS 19-03. Scope includes completion of Preliminary Map Products, distribution of preliminaries, Flood Risk Products upload (CSLF, FDAG, FRA, FRD), and all Post-Preliminary Map Production elements (CCO meeting, community feedback, due process/appeals, final FIRM production and distribution). Includes a minor redelineation of Spoon River Zone A (0.4 mi) with LiDAR from ISGS. Topographic and hydrologic/hydraulic data are from prior MAS ISWS 19-03.
 - **Deliverables:** Complete set of Preliminary FIRM panels; FIS Report; Preliminary SOMA; Flood Risk Products (FRD, CSLF); CCO meeting materials; due process documentation; final effective FIRM panels and distribution.
 - **Period:** 09/01/2021 – 09/30/2025
 - **Budget:** \$175,000
 - **Multi-year:** Yes — prior phase ISWS 19-03 covered data development through QR2; this MAS is the **final** phase, taking the project to effective. No additional FRPs expected after this.
 - **Recommended Next Steps:** Ensure community engagement is robust during 30-day CCO review period. Confirm no outstanding appeals that could delay effective date. This project should reach an effective FIRM within this grant period.
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ISWS 21-06: Montgomery County — Data Development (New Zone A Study)

- **Activity Type:** Flood Risk Project — Data Development (New Zone A hydraulics and mapping)
- **Phase:** Phase 1 (first MAS for Montgomery County); new study initiation
- **Scope:** Performs full H&H data development for Montgomery County streams. Hydrology: USGS Regional Regression Equations for ~248 sq mi of county study streams; HEC-HMS for five lakes (Lake Lou Yaeger, Lake Glenn Shoals, Lake Hillsboro, Coffeen Lake, Five Mile Lake). Hydraulics: Option B (simplified analysis) for 261.4 total miles of streams, producing Zone A approximate mapping (no BFE/detailed studies). Includes levee identification for hydraulically significant levees, Project Initiation Community Coordination (PICC), project outreach/communication plan, base map (IDOT aerial), and topographic data. No Discovery, no Flood Risk Products, no mapping delineation table entries—data development deliverables go to QA/QC and then presumably a future FIRM database and preliminary map MAS.
- **Deliverables:** H&H reports (hydrology and hydraulics) with independent QA/QC review notes; topographic dataset; base map; project management documentation; communication plan; PICC meeting materials.
- **Period:** 09/01/2021 – 09/30/2025
- **Budget:** \$400,000
- **Multi-year:** Yes. This is Phase 1 / Data Development only. Future MASs will be required for: floodplain mapping/delineation, Draft FIRM Database, Preliminary Map Production, and Post-Preliminary Map Production.

- **Recommended Next Steps:** After H&H QA/QC is complete, fund a continuation MAS for floodplain delineation and Draft FIRM Database. Plan for multiple additional grant years to reach effective FIRM.
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ISWS 21-07: Hamilton County — Data Development (New Zone A Study)

- **Activity Type:** Flood Risk Project — Data Development (New Zone A hydraulics and mapping)
 - **Phase:** Phase 1 (first MAS for Hamilton County); new study initiation
 - **Scope:** Performs full H&H data development for Hamilton County streams. Hydrology: USGS Regional Regression Equations for most streams; HEC-RAS 2D "Rain on Grid" for Skillet Fork Tributaries (165 sq mi 2D analysis). Hydraulics: Option B for 276 miles total plus 165 sq mi 2D area — all Zone A approximate mapping. Includes levee identification, monitoring of new LiDAR availability, PICC meeting, project outreach/communication plan, and topographic data. No Discovery, no Flood Risk Products. Same structure as Montgomery County MAS.
 - **Deliverables:** H&H reports (hydrology and hydraulics) with independent QA/QC; topographic dataset; base map; project management documentation; communication plan; PICC meeting materials.
 - **Period:** 09/01/2021 – 09/30/2025
 - **Budget:** \$400,000
 - **Multi-year:** Yes. Data Development only; identical lifecycle gap as Montgomery County. Future MASs needed for delineation, FIRM database, and Preliminary/Post-Preliminary map production.
 - **Recommended Next Steps:** Confirm LiDAR availability with ISGS as it will affect delineation quality. After H&H QA/QC passes, fund a continuation MAS for floodplain delineation and Draft FIRM Database. Identify levee situations early to avoid late-stage delays.
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ISWS 21-08: Middle and Lower Wabash HUC8 — Data Development (Continuation)

- **Activity Type:** Flood Risk Project — Data Development (continuation of ISWS 18-03)
- **Phase:** Phase 2+ (supplements ISWS 18-03 FY18 data development, cross-state project)
- **Scope:** Continues work in the Middle Wabash-Busseron (HUC 05120111) and Lower Wabash (HUC 05120113) watersheds. FY21 funding supplements ISWS 18-03 to complete hydraulic analysis tasks: adds levees to the Wabash River hydraulic model; develops natural valley and constricted floodway/floodplain delineation scenarios for (1) Wabash River main stem, (2) select Indiana tributaries, and (3) levees; incorporates additional survey data; performs additional analysis for 4%, 0.2%, and 1%+ flood events. White River hydraulics (Option E) also included. Involves coordination with Indiana project partners and FEMA RSC for cross-state components. Independent QA/QC of Wabash River model included.
- **Deliverables:** Updated Wabash River hydraulic model with levees; floodway/floodplain analysis scenarios; additional event frequency analyses; independent QA/QC report; MIP task documentation.
- **Period:** 09/01/2021 – 09/30/2025
- **Budget:** \$150,000
- **Multi-year:** Yes. Explicitly states: "ISWS will request funding when MAS is prepared for QR-3 to Effective." Prior phase: ISWS 18-03. The new floodway establishment will have significant community impact.

- **Recommended Next Steps:** Upon completion of levee modeling and floodway scenarios, fund a Preliminary Map Production MAS (QR3 to Effective). Advance cross-state coordination with Indiana to ensure alignment on shared flooding source delineations. Engage levee sponsors early regarding PAL/accreditation implications of the new floodway analysis.
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ISWS 21-09: Peoria County — Community Engagement Supplement & Spoon River Redelineation

- **Activity Type:** Flood Risk Project — Supplemental Engagement and Redelineation
 - **Phase:** Supplement 3 (third MAS for Peoria County; prior MASs: ISWS 18-04 and ISWS 20-13)
 - **Scope:** Supplements prior Peoria County MASs. Provides continued levee engagement, outreach, and timely technical response to stakeholder questions as the project advances through its lifecycle. Redelineates Spoon River Zone AE (10.3 mi) using existing LiDAR. Monitors potential changes in ground conditions, culverts, bridges, and other structures. Continues levee-related research and information provision to stakeholders.
 - **Deliverables:** Community outreach documentation; Spoon River floodplain mapping redelineation (10.3 mi AE); technical response documentation; levee engagement records.
 - **Period:** 09/01/2021 – 09/30/2025
 - **Budget:** \$75,000
 - **Multi-year:** Yes. Third supplement to an ongoing multi-year project. Main project (ISWS 18-04) is presumably in or approaching Preliminary/Post-Preliminary stage; this MAS provides bridge funding for engagement and a specific redelineation gap.
 - **Recommended Next Steps:** Identify what stage the main Peoria County FIRM project is at and whether a Preliminary Map Production MAS is already funded or needed. Resolve outstanding levee accreditation/PAL issues. This ongoing engagement pattern suggests levee complexity is the primary bottleneck.
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ISWS 21-10: Bond County — Data Development Supplement (Keyesport Levee Support)

- **Activity Type:** Flood Risk Project — Data Development Supplement (supports ISWS 19-07 Paper Inventory Reduction)
- **Phase:** Supplemental data development phase; prior MAS: ISWS 19-07 Bond County
- **Scope:** Provides supplemental data development funding to support the existing Bond County Paper Inventory Reduction project (ISWS 19-07). Key technical issue is the **Keyesport Levee**, which is on a path toward accreditation submittal — the outcome of the accreditation process will determine the ultimate flood mapping for Keyesport and potentially hold up the FIRM. Hydraulics includes Carlyle Lake backwater analysis affecting Spring Branch, Keyesport Branch, and Carlyle Lake Trib H backwater areas. Standard project management is included.
- **Deliverables:** Hydraulic analysis for Carlyle Lake/Keyesport area streams; levee coordination documentation; project management documentation.
- **Period:** 09/01/2021 – 09/30/2025
- **Budget:** \$50,000

- **Multi-year:** Yes. Supplements ISWS 19-07. Future mapping deliverables (FIRM database, preliminary) are contingent on Keyesport Levee accreditation outcome.
- **Recommended Next Steps:** Pursue Keyesport Levee accreditation submittal to USACE/FEMA. Until levee status is resolved (accredited or not), the FIRM mapping for Keyesport area cannot be finalized. Fund a follow-on MAS for FIRM database and Preliminary Map Production once levee status is determined.

ISWS 21-11: COMS SOW — Statewide Community Outreach and Mitigation Strategies

- **Activity Type:** Community Outreach and Mitigation Strategies (COMS SOW)
- **Phase:** FY21 standalone 2-year COMS grant
- **Scope:** Statewide programmatic outreach and mitigation strategy activities. Key tasks include: (1) COMS Engagement Plan development — standalone plan to be integrated into future joint Business Plan with IDNR/OWR (\$20,000); (2) Strategic Planning for Community Engagement — awareness and action strategies for prioritized communities (\$25,000); (3) Digital presence maintenance — ISWS website and stakeholder information systems (\$20,000 implied); (4) SAFR (Stakeholder Assessment of Flood Risk) user guide development — how-to document for navigating the website and applying flood risk assessment data (\$25,000); (5) Mentoring and best practices — participation in CTP Mentoring Pilot Program, sharing GIS/Python tools, 1–2 best practice write-ups (\$55,000).
- **Deliverables:** COMS Engagement Plan (within 6 months of award); strategic community engagement documentation; updated website content; SAFR user guide; mentoring reports and best practice documents.
- **Period:** 09/01/2021 – 09/30/2023
- **Budget:** \$125,000
- **Multi-year:** N/A (COMS SOWs are biennial; not a multi-phase mapping project)
- **Recommended Next Steps:** Integrate COMS Engagement Plan into the joint ISWS/IDNR-OWR Business Plan in a future funding year. Continue digital tools development. Publish SAFR user guide and distribute to communities.

ISWS 21-12: Kishwaukee HUC8 Watershed — Data Development Phase 2

- **Activity Type:** Flood Risk Project — Data Development (Phase 2 of 2-phase data development)
- **Phase:** Phase 2 (Phase 1: ISWS 20-12 FY20, which covered topographic data and base map)
- **Scope:** Performs hydraulic (and hydrology) data development for the Kishwaukee HUC8 watershed as Phase 2 following Phase 1 (ISWS 20-12). Topographic data and base map are explicitly NOT in this MAS (covered by ISWS 20-12). Hydraulics for Kishwaukee River: Option D, 17.4 miles, Zone AE with BFE and Floodway, 1D modeling. Levee identification included. Covers a large multi-county watershed (Winnebago, Boone, McHenry, DeKalb, Kane, Ogle) with major communities including Rockford, Belvidere, Woodstock, Marengo, Huntley, Algonquin.
- **Deliverables:** H&H reports (hydrology and hydraulics) with independent QA/QC; levee identification documentation; project management records; communication plan. Note: FIRM database and mapping delineation are not shown in this MAS — those will require a future MAS.
- **Period:** 09/01/2021 – 09/30/2025

- **Budget:** \$750,000
- **Multi-year:** Yes. Phase 2 of a 2-phase data development sequence (ISWS 20-12 → ISWS 21-12). Future MASs needed for: floodplain delineation, Draft FIRM Database, Preliminary Map Production, and Post-Preliminary production. Given the size of the watershed, multiple future phases are likely.
- **Recommended Next Steps:** After H&H QA/QC is complete, fund Phase 3 for floodplain delineation and Draft FIRM Database. The Kishwaukee watershed is one of the larger and more complex projects in the portfolio; plan for 2+ additional grant years to reach effective FIRM.

Cross-Cutting Observations

1. Mississippi River Corridor — Three Co-Occurring PMR Projects (21-02, 21-03, 21-04)

Rock Island, Whiteside, and Henry Counties are all advancing Mississippi River-adjacent PMRs using data from the same prior grant (MAS ISWS 17-06). These three projects share the same FEMA RPO, same period of performance, and all stop at QR1 FIRM Database. They should be coordinated for a unified Preliminary Map Production cycle to maximize efficiency and reduce community confusion. Whiteside and Henry are lower-budget (\$150K and \$125K) compared to Rock Island (\$770K), reflecting Rock Island's more complex technical scope (new H&H, 2D modeling, levee analysis).

2. Multi-Year Data Development Pattern — Projects Needing Future Phases

Five projects are explicitly "data development only" with no pathway to final FIRM maps within this grant:

- **ISWS 21-06 (Montgomery)** — needs delineation, FIRM database, Preliminary, Post-Preliminary
- **ISWS 21-07 (Hamilton)** — same lifecycle gap
- **ISWS 21-08 (Wabash)** — explicitly flags "will request funding for QR-3 to Effective"
- **ISWS 21-12 (Kishwaukee)** — Phase 2 of data development; delineation and FIRM database still needed

These represent a significant budget investment in FFY21 (\$1.7M combined) that will not yield final FIRM products without continued future funding. Planning ahead for FY23/FY25 MASs for these projects is important.

3. Levee Issues as a Cross-Cutting Bottleneck

Five projects involve levee complications that could delay final FIRM issuance:

- **Rock Island (21-02)** — 7 levee areas, PAL status pending H&H outcome
- **Wabash (21-08)** — cross-state levee modeling; "new floodway will have significant impact"
- **Peoria (21-09)** — ongoing multi-year levee engagement (3rd supplement MAS)
- **Bond (21-10)** — Keyesport Levee accreditation outcome controls mapping for entire community
- **Hamilton (21-07)** — levee identification ongoing

Levee accreditation timelines (USACE/FEMA) are often outside ISWS's control and represent the primary schedule risk across the portfolio.

4. Recurring Watersheds Suggest a Multi-Cycle Program

Several watersheds appear in prior-year MASs that are being carried forward:

- **ISWS 17-06** feeds Rock Island, Whiteside, and Henry PMRs in FY21
- **ISWS 18-03** continues into Wabash FY21
- **ISWS 18-04 / 20-13** continues into Peoria FY21 (3rd supplement)
- **ISWS 19-03** is completed by Stark FY21
- **ISWS 19-07** is supplemented by Bond FY21
- **ISWS 20-12** (Phase 1) is continued by Kishwaukee Phase 2 FY21

This demonstrates a 4–8+ year lifecycle from initial grant to effective FIRM for most studies, consistent with FEMA's Risk MAP program expectations.

5. Stark County (21-05) — Lone Project Reaching Final FIRM

Of all 9 FRP MASs, only **Stark County (ISWS 21-05)** is scoped to reach an effective FIRM within this grant period. This project completes the full lifecycle (Preliminary → Distribution → CCO → Due Process → Effective) and should be tracked as the portfolio's near-term "completion" project.

6. Kishwaukee — Largest Single FRP (\$750K) and Most Complex Future Need

The Kishwaukee HUC8 Phase 2 is the largest single FRP investment at \$750,000 and covers a major urban/suburban watershed (Rockford metro, McHenry/Kane collar counties). Given the population served, the complexity of the hydraulic analysis, and the certain need for 2+ additional future MASs, this project should receive attention in long-range grant planning.